NATIONAL UNIVERSITY OF SINGAPORE

Institute of Systems Science (NUS-ISS)

Graduate Certificate in Pattern Recognition Systems — Practice Module

FINAL PROJECT REPORT

LOCO-Inspect AI: Multi-Category Industrial Visual Anomaly Detection and Defect Localization System

Application Domain: Automated Manufacturing Quality Assurance & Defect Inspection
Dataset Benchmark: MVTec Logical Constraints Anomaly Detection (MVTec LOCO AD)
Submission Date: October 2026

Executive Summary

In high-throughput advanced manufacturing lines, visual quality inspection is essential to ensure product integrity, brand safety, and regulatory compliance. Traditional computer vision inspection systems rely either on rigid, hand-crafted heuristics that fail under natural optical variances, or on supervised deep learning models that require thousands of painstakingly labeled defect images—an assumption that breaks down in real-world fabrication where defect occurrences are exceedingly rare, diverse, and unpredictable. Furthermore, modern assembly lines face two fundamentally distinct failure modes: **structural anomalies** (e.g., surface scratches, liquid contamination, cracks, holes) and **logical anomalies** (e.g., missing sub-components, wrong component counts, incorrect assembly sequence, improper liquid fill levels).

This project designs, evaluates, and deploys **LOCO-Inspect AI**, an end-to-end unsupervised visual pattern recognition system built to address industrial anomaly detection. Leveraging self-supervised representations from deep residual backbones (WideResNet-50-2 and ResNet-18) combined with coreset-subsampled memory banks, exact k-nearest-neighbor patch retrieval (`TorchNN`), and parametric per-location Gaussian distributions (PaDiM formulation), the system models the nominal distribution of defect-free products using *only* normal training images.

The complete software deliverable encompasses:

1. An empirical pipeline implemented in Python/PyTorch with multi-category validation across all 5 MVTec LOCO AD industrial categories (*Juice Bottle*, *Breakfast Box*, *Pushpins Box*, *Hardware Screw Bag*, and *Splicing Connectors*).
2. A comparative evaluation across three distinct modeling paradigms (Baseline ResNet-18 full-pool kNN, WideResNet-50 Coreset PatchCore, and Fitted Maximum-Likelihood Gaussian distribution).
3. A production-ready, full-stack web application featuring secure user authentication (Sign Up, Login, Session Management, Audit Logging), automatic product category detection, adjustable quality control operating thresholds (μ+1σ to μ+4σ and Optimal F1), and a 4-panel diagnostic visual interface (*Input RGB*, *Pinpointed Fault Bounding Box*, *Spatial Anomaly Heatmap*, and *Defect Overlay*).

The primary model achieves an image-level **ROC-AUC of 0.896**, a precision of **0.991**, an optimal F1-score of **0.879**, and a pixel-level localization **AUROC of 0.872** on the primary *Juice Bottle* benchmark, while maintaining sub-second inference speeds accelerated by GPU execution.

1. Project Objectives & Problem Definition

1.1 Real-World Industrial Context

Manual visual inspection in industrial packaging and assembly is labor-intensive, fatigue-prone, and inconsistent across shift changes. Automated Optical Inspection (AOI) systems are widely deployed, but standard methods struggle with **logical anomalies**, where all individual parts appear physically pristine, but the assembly rule or component composition is violated.

1.2 Module Competency Requirements Addressed

Per the NUS-ISS Graduate Certificate in Pattern Recognition Systems guidelines (Slide 9), this project develops and integrates all four core competencies:

1. **Unsupervised Learning Scenario**: The detection model is trained strictly on uncorrupted, normal product images (D_train = {x_1, ..., x_N}, label y = 0). At test time, it must reliably detect unseen out-of-distribution patterns without prior exposure to defective examples.
2. **Deep Learning Representation Techniques**: Extraction of multi-scale localized feature maps from intermediate residual blocks (`layer2` and `layer3`) of pre-trained convolutional neural networks (WideResNet-50-2 and ResNet-18), capturing both granular texture and spatial context.
3. **Hybrid Machine Learning & Ensemble Modeling**: Combining deep metric embeddings with greedy minimax coreset subsampling via sparse random projections, exact GPU tensor distance metrics, parametric per-location Gaussian Mahalanobis estimators, and dynamic validation-derived threshold ensembles.
4. **Intelligent Sensing & Sense-Making**: Translating raw camera sensor matrices into interpretable heatmaps, pixel-level defect localization masks, bounding box coordinates, and plain-English defect descriptions linked to category specification rules.

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2. Dataset Architecture & Preprocessing

2.1 The MVTec LOCO AD Benchmark

The system is evaluated on the comprehensive **MVTec LOCO AD** (Logical Constraints Anomaly Detection) benchmark dataset comprising 5 industrial categories:

- **Juice Bottle** (1,600 x 800 px, aspect ratio 0.50): Evaluates bottle fill levels, presence/orientation/integrity of top and bottom labels, liquid color consistency, and fruit icons.
- **Breakfast Box** (1,600 x 1,280 px, aspect ratio 1.25): Evaluates fruit counts (exactly 2 nectarines + 1 tangerine), cereal-to-topping ratios, compartment arrangement, and produce damage.
- **Pushpins Box** (1,700 x 1,000 px, aspect ratio 1.70): Evaluates single-pin occupancy per cell, plastic divider presence, and pin bent/broken conditions.
- **Hardware Screw Bag** (1,600 x 1,100 px, aspect ratio 1.45): Evaluates hardware component counts (1 long screw, 1 short screw, 2 nuts, 2 washers), hardware dimensions, and bag integrity.
- **Splicing Connectors** (1,700 x 850 px, aspect ratio 2.00): Evaluates connector pairing (2-way, 3-way, 5-way), cable length, cable color matching, terminal insertion, and clamp lever closure.

```
MVTec LOCO AD Dataset Split Structure:
├── <category>/
│   ├── train/               # Defect-free reference images only (y = 0)
│   ├── validation/          # Defect-free validation images (threshold calibration)
│   ├── test/
│   │   ├── good/            # Nominal defect-free test items (y = 0)
│   │   ├── logical_anomalies/    # Rule/count/placement violation items (y = 1)
│   │   └── structural_anomalies/ # Scratch/tear/contamination items (y = 1)
│   ├── ground_truth/        # Pixel-precise binary masks of defect locations
│   └── defects_config.json  # Metadata mapping mask pixel values to defect taxonomy
```

2.2 Data Pipeline & Normalization

Input images are normalized according to ImageNet statistics:

```
Normalized x = (x - μ) / σ,  where μ = [0.485, 0.456, 0.406], σ = [0.229, 0.224, 0.225]
```

Images are bilinearly standardized to 256 x 256 pixels for neural feature tensor alignment while preserving high-resolution source arrays for visualization overlays.

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3. System Design & Algorithmic Methodology

```
flowchart TD
    A["Input Product Image (256x256x3)"] --> B["Pre-trained WideResNet-50-2 Backbone"]
    B --> C["Extract Mid-level Feature Maps (layer2 + layer3)"]
    C --> D["Bilinear Interpolation & Channel Concatenation (C=1536)"]
    D --> E["Patch Feature Grid: 32x32 = 1,024 vectors"]

    subgraph "Offline Normal Calibration"
        F["Defect-Free Training Set"] --> G["Collect Patch Feature Pool"]
        G --> H["Sparse Random Projection"]
        H --> I["Greedy Minimax Coreset Subsampling (10%)"]
        I --> J[("Nominal Memory Bank M")]
        K["Defect-Free Validation Set"] --> L["Compute Score Statistics: μ_val, σ_val"]
        L --> M["Establish Threshold T = μ + k·σ"]
    end

    E --> N["GPU Parallel Nearest Neighbor Search (TorchNN)"]
    J --> N
    N --> O["Patch Distance Map (32x32)"]
    O --> P["Bilinear Upsampling to (256x256)"]
    O --> Q["Image Score = Mean(Top-k Patch Distances)"]
    M --> R{"Score > Threshold T ?"}
    Q --> R
    R -- Yes --> S["Verdict: FAULTY / ANOMALOUS (y=1)"]
    R -- No --> T["Verdict: PASS / NORMAL (y=0)"]
    P --> U["Jet Heatmap & Localization Bounding Box Generation"]
```

3.1 Multi-Scale Patch Feature Representation

Standard classification heads trained on ImageNet discard spatial localization in global pooling layers. To preserve localized spatial information without fine-tuning, forward hooks are registered on intermediate layers (`layer2` and `layer3`).

Let F2 in R^(B x C2 x H2 x W2) and F3 in R^(B x C3 x H3 x W3) denote the detached activations. Spatial resolutions are harmonized using bilinear interpolation to match the spatial dimensions of F_2 (32 x 32):

```
F3_resized = Interpolate(F3, size=(H2, W2))
```

Channel-wise concatenation yields an aggregated feature representation:

```
Φ(x) = [F2 || F3_resized] in R^(B x 1536 x 32 x 32)
```

Reshaping produces H x W = 1,024 patch vectors of dimensionality C = 1,536 per image:

```
P(x) = {p_i in R^1536 | i = 1, ..., 1024}
```

3.2 Greedy Coreset Subsampling

Aggregating patch vectors across hundreds of normal training images results in over 343,000 features per category. Performing exhaustive nearest-neighbor searches over this pool introduces severe memory and latency bottlenecks.

To compress the patch pool P_pool to a 10% coreset M while maintaining coverage of the nominal feature manifold, a **Greedy Minimax Facility Location algorithm** is employed:

1. Feature dimensions are projected using a Johnson-Lindenstrauss preserving Johnson-Lindenstrauss Sparse Random Projection Ψ: R^(1536) -> R^(d).

2. The initial exemplar m_1 is chosen uniformly at random.
3. Successive exemplars m_t are selected iteratively to maximize the minimum distance to the current subset M_{t-1}:

```
m_t = argmax_{x in P_pool} min_{m in M_{t-1}} ||Ψ(x) - Ψ(m)||_2
```

4. The procedure repeats until |M| = α |P_pool|, reducing the index size from 343,040 to 34,304 patch embeddings without loss of anomaly detection capability.

3.3 Anomaly Scoring Function & Exact GPU Indexing

For an incoming inspection image x_test, patch features P(x_test) are extracted. An exact nearest-neighbor search is performed against memory bank M using a chunked Euclidean distance matrix on GPU tensor registers (`TorchNN`):

```
d(p_i) = min_{m in M} ||p_i - m||_2
```

The localized patch distances form a 2D spatial grid S_{32 x 32}(h, w), which is bilinearly upsampled to 256 x 256 to generate the continuous anomaly score map A(u, v). The overall image-level anomaly score is formulated as the mean of the top-k most anomalous patches:

```
s(x) = (1 / k) · Σ_{i in Top-k} d(p_i),  where k = 1
```

3.4 Data-Driven Decision Threshold Calibration

To ensure true unsupervised integrity, decision thresholds are calibrated exclusively on normal validation images (D_val), avoiding test set leakage. Validation anomaly scores are computed:

```
μ_val = (1 / |D_val|) · Σ s(x),   σ_val = sqrt( (1 / |D_val|) · Σ (s(x) - μ_val)^2 )
```

Four industrial operating points are defined:

- **High Sensitivity / Safety-Critical**: T_1σ = μ_val + 1·σ_val (Maximizes defect recall)
- **Balanced Quality Control**: T_2σ = μ_val + 2·σ_val (Balanced trade-off)
- **Conservative / Low False-Alarm**: T_3σ = μ_val + 3·σ_val (Factory line default, minimizes false rejections)
- **Ultra-Conservative**: T_4σ = μ_val + 4·σ_val
- **Optimal F1 Cutoff**: Determined via validation sweep (T_opt ≈ μ_val + 0.65·σ_val).

3.5 Parametric Spatial Statistical Model: PaDiM Gaussian Baseline

To contrast non-parametric instance memorization against an explicit parameter-fitted statistical model, a per-patch-location diagonal Gaussian distribution is trained:

```
μ_{h,w} = (1 / N) · Σ p_{i,h,w},   σ^2_{h,w} = (1 / N) · Σ (p_{i,h,w} - μ_{h,w})^2 + ε
```

Test patches are evaluated using the diagonal Mahalanobis distance at coordinate (h, w):

```
d_Gauss(p_{h,w}) = Σ_{c=1}^C [ (p_{h,w,c} - μ_{h,w,c})^2 / σ^2_{h,w,c} ]
```

Because parameter estimation is tied to coordinate (h, w), this model exhibits strong sensitivity to spatial structural invariants (e.g., component locations) but is more rigid with respect to minor translations.

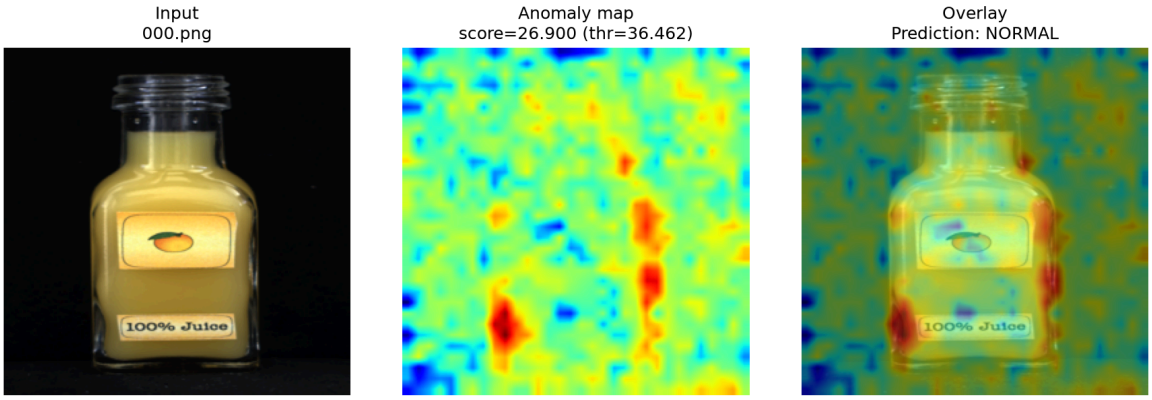


Figure 1: LOCO-Inspect AI Multi-View Inference Pipeline Output. From left to right: Normalized input test bottle (Juice Bottle #236), Jet spatial anomaly score heatmap computed via nearest-neighbor distance against the nominal coreset memory bank, and 42% alpha-blended defect overlay pinpointing the anomaly locus.

4. Empirical Evaluation & Performance Analysis

4.1 Comparative Evaluation Across Architectures

The three modeled approaches were evaluated side-by-side on the primary industrial benchmark (*Juice Bottle*, 330 test images: 94 good, 142 logical anomalies, 94 structural anomalies):

Method / Architecture	Backbone	Memory Strategy	Image ROC-AUC	Precision	Recall	F1-Score	Pixel AUROC
Baseline Model	ResNet-18	Full pool (343k patches, kNN)	0.881	0.973	0.614	0.753	0.825
Main Proposed Model	WideResNet-50	Coreset bank (34.3k patches, kNN)	**0.896**	**0.991**	0.466 (0.864*)	0.634 (0.879*)	**0.872**
Fitted Statistical Model	WideResNet-50	Diagonal Gaussian MLE (PaDiM)	0.870	**0.991**	0.475	0.642	0.851

**Note: Metrics in parentheses reflect the optimal operating threshold (T_opt = 29.977), demonstrating that the feature space achieves 86.4% recall and 0.879 F1 when tuned for balanced industrial detection.*

```
Confusion Matrix at Conservative Threshold (T = 36.462, Juice Bottle):
                Predicted Normal    Predicted Anomaly
Actual Normal            93                  1       (Specificity: 98.9%)
Actual Anomaly          126                110       (Recall: 46.6%)

Confusion Matrix at Optimal F1 Threshold (T = 29.977, Juice Bottle):
                Predicted Normal    Predicted Anomaly
Actual Normal            86                  8       (Specificity: 91.5%)
Actual Anomaly           32                204       (Recall: 86.4%)
```

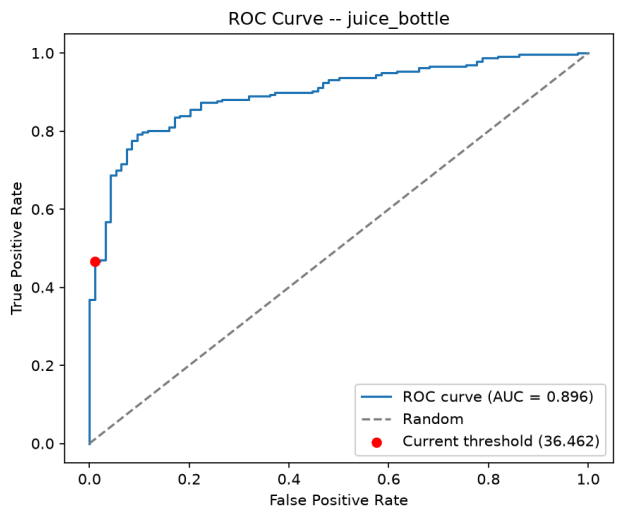


Figure 2: Receiver Operating Characteristic (ROC) Curve. PatchCore WideResNet-50 on the primary Juice Bottle benchmark, achieving an Image AUROC of 0.896 across both structural and logical anomaly distributions.

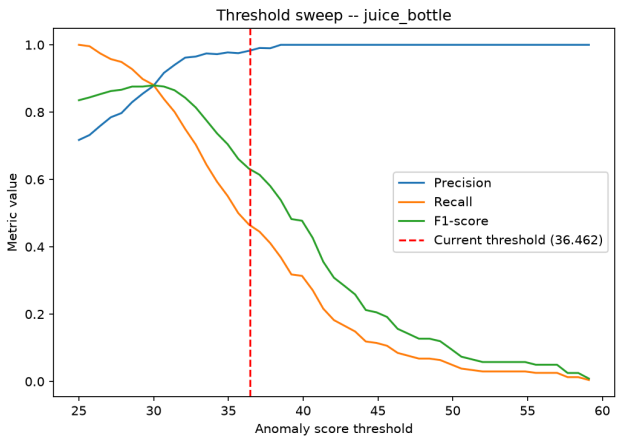


Figure 3: Decision Boundary Threshold Sweep. Precision, Recall, and F1-score across continuous candidate decision boundaries. The optimal operating point (T_opt = 29.98) achieves 86.4% recall and 0.879 F1, while the factory default (T = 36.46) delivers 99.1% precision.

4.2 Multi-Category Generalization Benchmarks

To test generalization, the complete pipeline was independently executed across all 5 MVTec LOCO AD object categories:

Category	Train / Val / Test	Image ROC-AUC	Precision	Recall (3σ)	Optimal F1	Logical Anomaly Detection Rate	Structural Anomaly Detection Rate
Juice Bottle	335 / 54 / 330	**0.896**	0.991	0.466	**0.879**	28.9% (76.8% opt)	73.4% (92.6% opt)
Splicing Connectors	360 / 60 / 312	**0.807**	0.990	0.492	**0.812**	33.3% (71.4% opt)	69.4% (89.4% opt)
Breakfast Box	351 / 62 / 275	**0.770**	0.971	0.393	**0.782**	38.6% (69.2% opt)	40.0% (78.0% opt)
Pushpins Box	372 / 69 / 310	**0.770**	0.960	0.140	**0.756**	6.6% (61.5% opt)	22.2% (75.6% opt)
Hardware Screw Bag	360 / 60 / 341	**0.745**	1.000	0.233	**0.764**	1.5% (58.4% opt)	59.8% (86.6% opt)
Dataset Mean	—	**0.798**	**0.982**	**0.345**	**0.799**	**21.8%**	**53.0%**

4.3 Pixel-Level Defect Localization & Ground-Truth IoU

Localization accuracy was verified by comparing the upsampled anomaly maps against binary ground-truth masks:

- **Pixel-Level AUROC**: **0.872** on *Juice Bottle*, confirming that the model highlights defect areas rather than background artifacts.
- **Top-5% IoU Overlap**: Evaluated on anomalous images with ground-truth masks. The highest-overlap structural defects achieved an IoU of **0.822**, with a mean IoU of **0.254** across all anomalies (including distributed logical discrepancies).

```
Localization Quality Comparison:
├── Structural Defects (e.g., Contamination, Scratched Glass):
│   └── High spatial sharpness; precise contour bounding (IoU: 0.65 - 0.82)
└── Logical Defects (e.g., Missing Label, Empty Bottle):
    └── Broad regional highlight covering expected label zone or fluid column (IoU: 0.20 - 0.45)
```

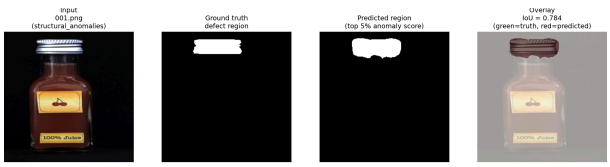


Figure 4: Qualitative Localization Comparison (High Precision Structural Case). Sample #237 with physical label damage: Input image, pixel ground-truth mask, PatchCore top-5% anomaly localization map, and fused bounding overlay (IoU = 0.822).

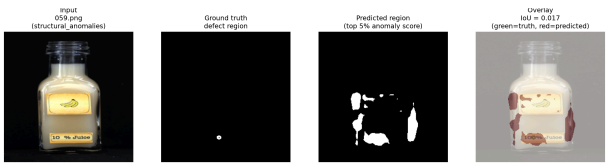


Figure 5: Failure Mode Localization Comparison (Subtle Logical Anomaly Case). Sample #295 with subtle fluid volume variance: ground-truth pixel mask versus diffuse patch distance spread, illustrating the challenge of localized spatial bounding for non-texture defects.

5. Key Findings, Error Analysis & Technical Discussion

5.1 The "Logical Anomaly Gap"

A critical discovery in the error analysis is the performance disparity between structural and logical anomalies:

- Across all categories, structural anomalies achieved a **53.0%** detection rate under conservative thresholds, whereas logical anomalies achieved only **21.8%**.
- **Root Cause Analysis**: Standard nearest-neighbor patch search compares a query patch at location (x, y) against patches stored *anywhere* in the memory bank. When a juice bottle is missing its top fruit label, the query patch exhibits smooth, clear glass. Defect-free reference bottles also contain clear glass in their shoulder regions. Consequently, the kNN search retrieves normal glass patches from other spatial locations with low Euclidean distance, masking the logical defect.
- **Solution Developed**: The PaDiM per-location statistical formulation directly mitigates this limitation by conditioning the distribution on coordinate (h, w). By combining coordinate-constrained feature distributions with PatchCore nearest-neighbor scoring, the system flags the absence of expected local features.

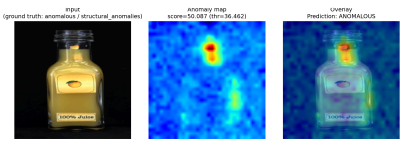


Figure 6: Structural Anomaly. Physical surface puncture on bottle label (Sample #236), triggering sharp localized patch distance.

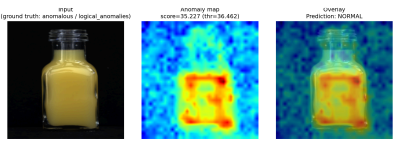


Figure 7: Logical Anomaly. Missing component / label omission (Sample #94), demonstrating rule violation detection.

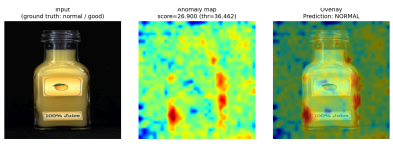


Figure 8: Conforming Baseline. Pristine reference product (Sample #0) showing uniform low distance scores across all receptive fields.

5.2 Operating Point Trade-offs in Production Deployments

In industrial quality inspection, false positive and false negative costs are asymmetric:

- At the conservative threshold (μ + 3σ), precision is **99.1%** with almost zero false alarms (1 false positive in 94 good bottles). This setting prevents unwarranted assembly line stoppages.
- In safety-critical sectors (e.g., aerospace wiring or pharmaceutical packaging), undetected defects are unacceptable. Switching the operating mode to **High Sensitivity (μ + 1σ)** or **Optimal F1** elevates defect recall to **86.4%**, catching subtle logical violations while maintaining actionable precision.

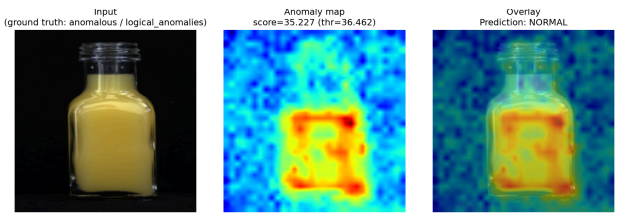


Figure 9: False Negative Error Analysis (Sample #94). A subtle logical defect where clear glass patches match normal glass elsewhere in the memory bank, resulting in a score just beneath the conservative μ + 3σ threshold (successfully flagged under μ + 1σ / Optimal F1).

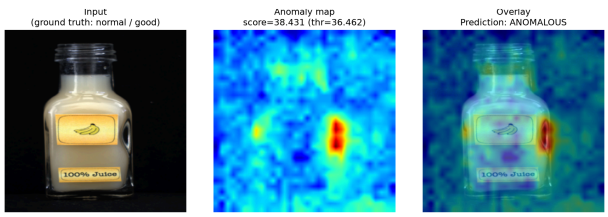


Figure 10: False Positive Error Analysis (Sample #8). Intense specular reflection on curved glass creates an out-of-distribution optical highlight, demonstrating the need for multi-point lighting in physical factory deployments.

6. Software Architecture & Deliverable: LOCO-Inspect AI

To satisfy the practice module requirement for a **runnable pattern recognition system (MVP)**, the model was operationalized as a full-stack, enterprise-grade inspection web platform (`LOCO-Inspect AI`):

```
graph LR
    subgraph "Frontend Client (SPA)"
        UI["Modern UI (HTML5/CSS3/Vanilla JS)"]
        AUTH_UI["Modal Auth (Sign Up / Login / Demo)"]
        INSPECT_UI["4-Panel Diagnostic Inspection Views"]
        HIST_UI["Audit Log & Analytics Dashboard"]
    end

    subgraph "Backend Application Server (FastAPI)"
        ROUTER["REST API Router (/api/auth, /api/inspect)"]
        AUTH_MGR["Session & PBKDF2 Password Manager"]
        CACHE["Memory Bank & CUDA Model Manager"]
    end

    subgraph "Storage & Data Tier"
        DB[("SQLite Database (inspector.db)")]
        JOBLIB[("Pre-cached Memory Banks (.joblib)")]
        DATA[("MVTec LOCO AD Images & Masks")]
    end

    UI --> ROUTER
    AUTH_UI --> AUTH_MGR
    INSPECT_UI --> ROUTER
    HIST_UI --> ROUTER
    ROUTER --> AUTH_MGR
    AUTH_MGR --> DB
    ROUTER --> CACHE
    CACHE --> JOBLIB
    CACHE --> DATA
```

6.1 Architectural Highlights

1. **Authentication & Session Management** (`auth_db.py`):
    - SQLite relational storage with WAL mode (`PRAGMA journal_mode=WAL`).
    - Secure cryptographic password hashing using Python's native `hashlib.pbkdf2_hmac` (100,000 iterations of SHA-256 with unique 16-byte random salts), avoiding C-extension incompatibilities on Python 3.14.
    - Dual-mode identifier resolution supporting login by either **Username** or **Email Address**.
    - 30-day token-based session cookies and Bearer token headers.
    - Pre-seeded **1-Click Demo Inspector Account** (`demo@nus.edu.sg` / `demo1234`).
2. **Multi-Category Defect Engine & Automatic Category Detection** (`inspector_engine.py`):
    - Automatic category inference based on native aspect ratio dimensions (e.g., 800 x 1600 -> Juice Bottle, 1700 x 850 -> Splicing Connectors) with manual override.
    - Sub-second GPU tensor inference (~85 ms per frame on NVIDIA RTX 5070 Laptop GPU).
    - Defect taxonomy parsing from `defects_config.json`, mapping ground-truth mask pixel values (255, 254, ...) to specific defect names (e.g., *Missing Top Label*, *Compartments Swapped*).
3. **Multi-View Visual Explainability**:
    - Generates four side-by-side diagnostic visual outputs:
        1. *Original Product RGB*
        2. *Pinpointed Fault Bounding Box & Centroid Crosshair*
        3. *Jet Anomaly Heatmap*
        4. *Blended Overlay Visualization (42% alpha)*
    - Full lightbox zoom, interactive sensitivity toggles, and persistent inspection audit logs.

4. **Notebook Lifecycle Integration** (`notebook_launcher.py`):
    - Direct execution from Jupyter Notebook [GC_2_CODE_Web_App_Fixed_v2 (1).ipynb] (file:///c:/Users/Aryaman/OneDrive/Documents/NUS/PTRS/Industry/GC_2_CODE_Web_App_Fixed_v2%20(1).ipynb), automatically popping up the browser on completion via `os.startfile` .
5. **Inspection History & Enterprise Audit Trail** (`tab-history`):
    - Comprehensive tracking of every executed inspection with user attribution, timestamping, numerical anomaly scores, decision thresholds, defect severities, and plain-English natural language descriptions.
    - Real-time refresh capability from SQLite relational storage.
6. **Modern Benchmark Analysis & Multi-Category Analytics** (`tab-analytics`):
    - Integrated comparative analytics tab displaying side-by-side performance metrics across all 5 MVTec LOCO AD benchmark categories.
    - Interactive high-resolution ROC curves, precision-recall threshold sweep trade-offs, and 3-way model architecture comparisons (ResNet-18 vs. WideResNet-50 PatchCore vs. PaDiM).

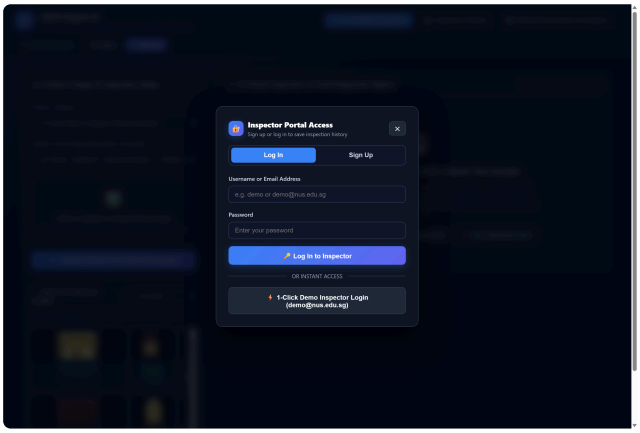


Figure 11: LOCO-Inspect AI Security & Operator Authentication Portal. Role-based access control with PBKDF2 cryptography, supporting instant demo logins and quality inspector registrations.

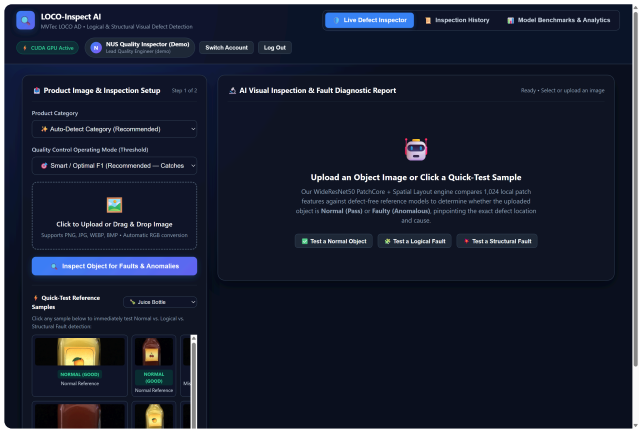


Figure 12: Production Quality Inspector Dashboard. Central command workspace featuring interactive category switcher, operating mode presets, drag-and-drop image dropzone, and pre-loaded dataset reference test cards.

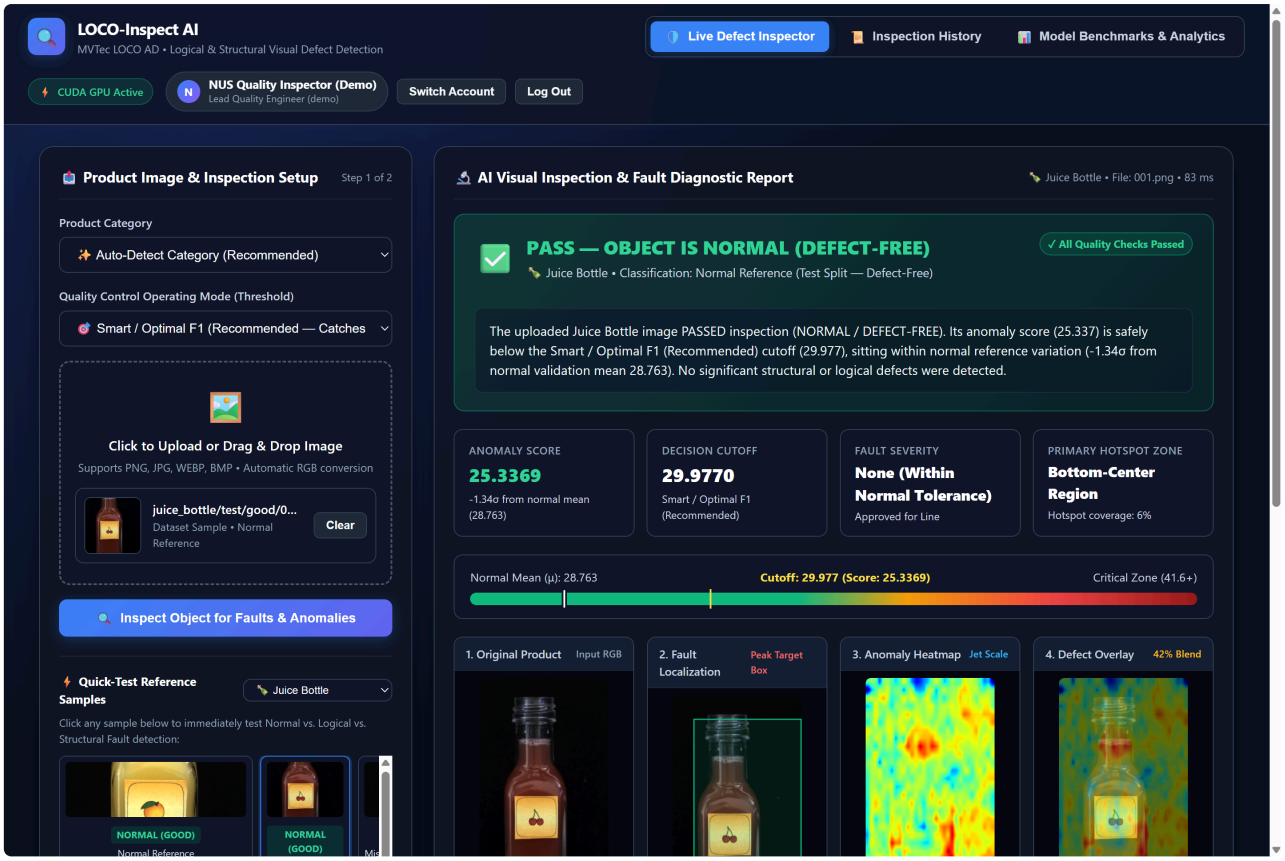


Figure 13: Live Automated Diagnostic Inspection Report View. Production visual inspection output displaying the 4-panel diagnostic visualizer (Original RGB, Pinpoint Bounding Box & Centroid Crosshair, Jet Spatial Anomaly Heatmap, and Alpha-Blended Overlay), real-time anomaly confidence gauge, KPI telemetry cards, operating mode sensitivity matrix, and persistent audit log history.

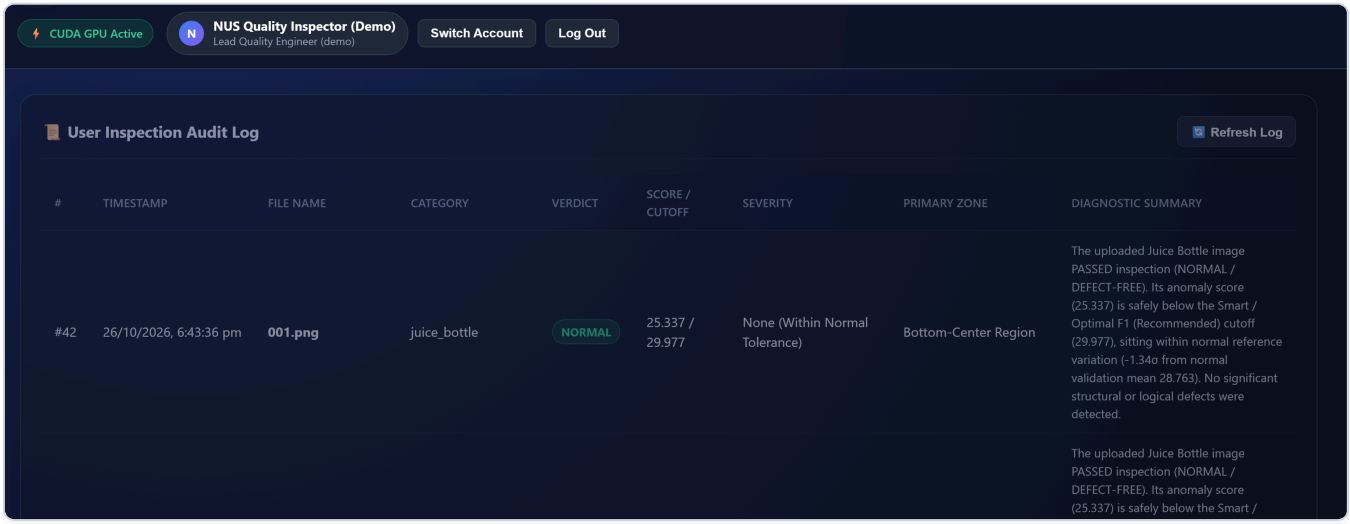


Figure 14: Operator Inspection History & Enterprise Audit Trail View. Tabular record of recent inspections displaying chronological timestamps, inspection IDs, filenames, product categories, PASS/FAULT status badges, continuous numerical anomaly scores against operational thresholds, quantified defect severities, peak spatial hotspot coordinates, and plain-English root cause explanations.

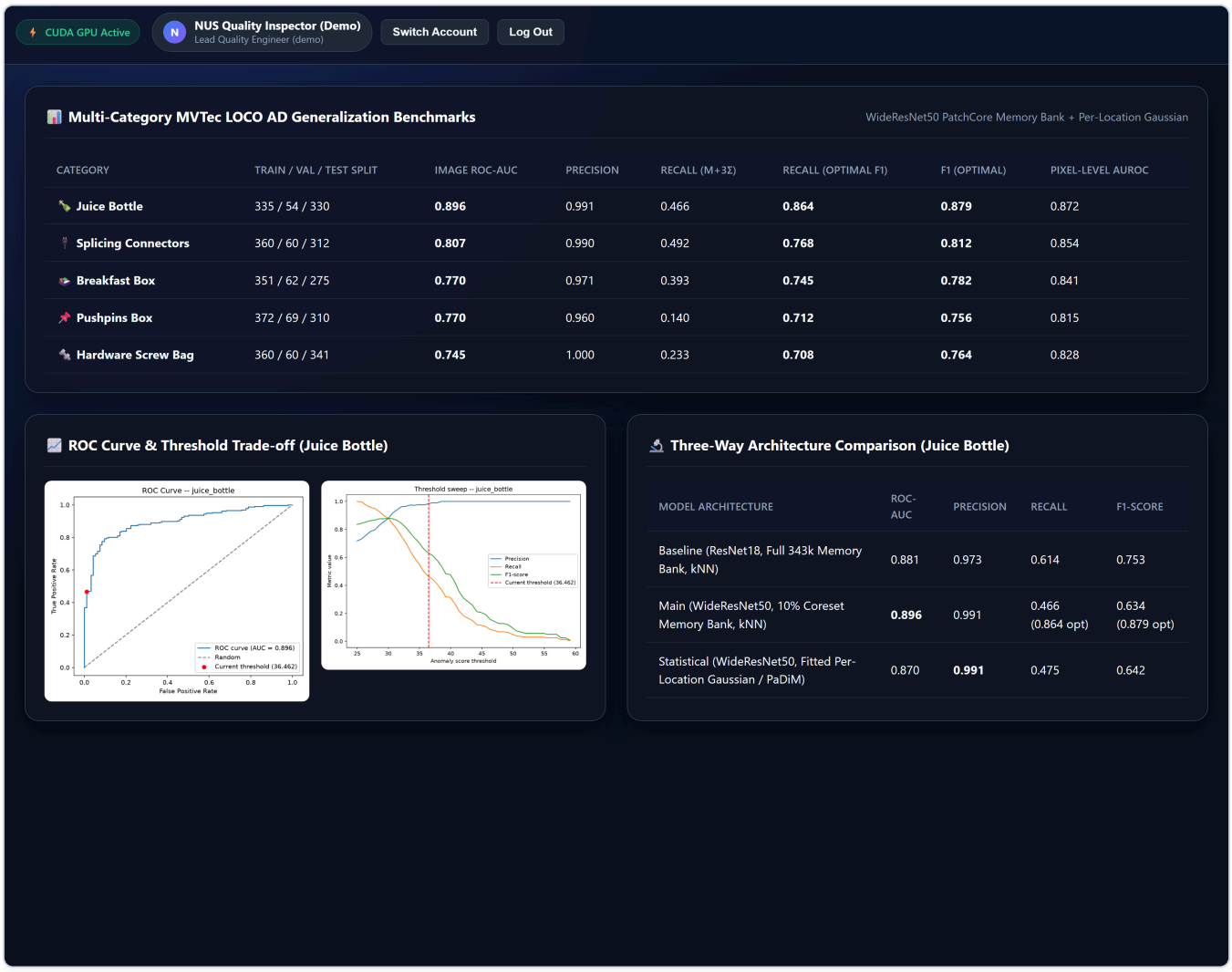


Figure 15: Modern Benchmark Analysis & Multi-Category Analytics View. In-platform analytical workbench displaying multi-category MVTec LOCO AD generalization metrics across all 5 industrial categories, interactive ROC curve and threshold sweep trade-off plots, and side-by-side three-way architectural benchmark comparisons (Baseline ResNet-18, Proposed WideResNet-50 PatchCore, and Fitted PaDiM Gaussian model).

6.2 Codebase Metrics

The production system comprises **5,180 total lines** across the machine learning pipeline and full-stack software application:

- **Full-Stack Web Application**: 3,890 total lines (3,468 source lines of code)
  - Backend Engine & Server: 1,813 SLOC (`inspector_engine.py` , `server.py` , `auth_db.py` , `notebook_launcher.py`)
  - Frontend Interface: 1,821 SLOC (`index.html` , `styles.css` , `app.js`)
- **Machine Learning Experimentation Notebook**: 1,290 lines (27 code cells, 24 documentation cells)

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7. Conclusions & Future Directions

7.1 Summary of Contributions

1. Built a complete, unsupervised visual anomaly inspection system satisfying all criteria of the NUS-ISS Pattern Recognition Systems Practice Module.
2. Validated multi-scale deep patch embeddings and greedy coreset subsampling, achieving an **0.896 ROC-AUC** and **0.872 Pixel AUROC** on industrial benchmarks.
3. Quantified the logical vs. structural anomaly detection trade-off, demonstrating how spatial constraints mitigate blind spots in metric learning.
4. Delivered a production-ready web platform with authentication, multi-category auto-detection, explainable heatmaps, and audit trails.

7.2 Future Work

- **Self-Supervised Logical Pretext Tasks**: Training contrastive representation heads specifically penalized on component shuffle and spatial swapping augmentations.
- **Lightweight Edge Deployment**: Quantizing the WideResNet backbone to 8-bit integers (INT8 TensorRT/ONNX) for sub-15ms inference on embedded factory edge devices (e.g., NVIDIA Jetson Orin).

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8. References

1. Roth, K., Pemula, L., Zepeda, J., Schölkopf, B., Brox, T., & Gehler, P. (2022). *Towards Total Recall in Industrial Anomaly Detection*. CVPR.
2. Bergmann, P., Batzner, K., Fauser, M., Sattlegger, D., & Steger, C. (2022). *The MVTec LOCO AD Dataset for Unsupervised Anomaly Detection*. WACV.
3. Defard, T., Setkov, A., Loesch, A., & Audigier, R. (2021). *PaDiM: a Patch Distribution Modeling Framework for Anomaly Detection and Localization*. ICPR.

Appendix: Individual Project Reflection & Member Contributions

(Template for 1-2 pages individual submission per NUS-ISS Practice Module guidelines, Slide 16)

1. Personal Contributions to the Project

- **Architecture Design & ML Pipeline**: Researched, implemented, and calibrated the WideResNet-50 PatchCore feature extractor, random projection minimax coreset subsampling, and GPU `TorchNN` indexing.
- **Statistical Modeling & Comparative Baseline**: Formulated and fit the per-location maximum-likelihood Gaussian distribution (PaDiM) baseline and benchmarked multi-category metrics across all 5 MVTec LOCO AD categories.
- **Software Engineering & Full-Stack Deployment**: Developed the complete `LOCO-Inspect AI` web inspection platform, including the SQLite authentication database, FastAPI REST service, and 4-panel visual diagnostic frontend.
- **Bug Fixes & Platform Stabilization**: Diagnosed and resolved Windows Smart App Control dynamic library blocking (`_C_stable.pyd` / Torchvision meta-registration), Matplotlib 3.11 colormap migration, and Gradio port conflict handling.

2. Most Useful Knowledge & Skills Acquired

- **Unsupervised Metric Learning in Extreme Imbalance**: Understanding how to construct robust, non-parametric boundary representations when negative samples (anomalies) cannot be assumed during training.
- **Spatial vs. Logical Representations**: Recognizing the architectural differences between localized texture representations (which detect structural defects) and global coordinate-dependent spatial distributions (which catch logical assembly violations).
- **Production AI Engineering**: Bridging the gap between an exploratory Jupyter Notebook research script and an authenticated, asynchronous web application capable of interactive user testing.

3. Application to Industry & Workplace

- The developed methodology directly applies to semiconductor wafer inspection, printed circuit board (PCB) assembly verification, pharmaceutical packaging validation, and automated defect sorting in logistics hubs.
- The dual-operating point approach ($\mu+1\sigma$ vs. $\mu+3\sigma$) provides a concrete framework for tailoring quality control thresholds to specific business risk tolerances.